DDI-Net 3.0

Explainable drug interaction intelligence

Multi-Drug Interaction Research Report

6 drugs · 15 pairs

DRUGS ANALYSED

1 **Warfarin**
DB00682 · approved

2 **Fluconazole**
DB00196 · approved, investigational

3 **Simvastatin**
DB00641 · approved

4 **Clarithromycin**
DB01211 · approved

5 **Amiodarone**
DB01118 · approved, investigational

6 **Digoxin**
DB00390 · approved

Report ID	DDI-20261015-045334-A6C3C708
Generated	15 October 2026, 04:53:34 (UTC, UTC+00:00)
Generated (UTC)	2026-10-15T04:53:34.361Z
Model	BIO-GINE M4 — frozen checkpoint bd45f84e3c1b2c33 per the site's frozen data; the multi-drug response itself does not name it
API report version	3.0
Analysis produced by the service	2026-10-15T04:53:34+00:00
Response received by the browser	2026-10-15T04:53:34.101Z
Service	http://127.0.0.1:8765
Language	English

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RESEARCH USE ONLY

DDI-Net is a research prototype — not a medical device and not clinical decision support. The absence of a documented interaction is not evidence of safety. The BIO-GINE score is a research signal of resemblance to documented pairs, not a probability of clinical harm.

1 Executive summary

- Drugs analysed: 6; unique pairs: 15. Each pair was examined on its own.

Pairs by kind of documentation found:

CI	Contraindication in a label	0
L	Named in a label	0
≠	Sources conflict	0
DB	Documented in a database	13
M	Model signal only	1
–	No documentation found — does not mean “safe”	1

- Pairs flagged by the service for a closer look: 14.
- Pairs whose own record the model was trained on (score not counted): 5.

> There is no overall score for the set, and none is computed: pairwise findings cannot be added up into a conclusion about several medicines together.

> Each pair is examined on its own. Pairwise findings say nothing about how three or more drugs behave together: no such model has been built or validated.

Uncertainty and evidence gaps

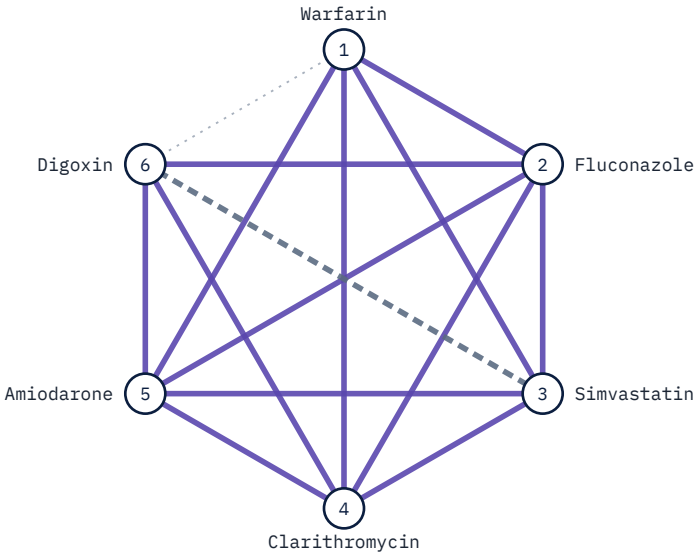
- Source “FDA labels (openFDA)”: unavailable. Its data is not part of this report.
- The service uses one model checkpoint (seed 0) and provides neither a per-pair confidence interval nor an ensemble estimate. Measured performance for this regime is given instead — an average over many pairs, not confidence in this pair.
- No source gives a severity, and none is assigned here.
- Doses, duration, age, kidney and liver function, other medicines and individual factors are not taken into account.

2 Overview of the drugs

#	INTERNATIONAL NAME (INN)	RUSSIAN NAME	DRUGBANK ID	DRUGBANK GROUPS	TYPE
1	warfarin	варфарин (transliteration)	DB00682	approved	small molecule
2	fluconazole	флуконазол (transliteration)	DB00196	approved, investigational	small molecule
3	simvastatin	симвастатин (transliteration)	DB00641	approved	small molecule
4	clarithromycin	кларитромицин (-transliteration)	DB01211	approved	small molecule
5	amiodarone	амиодарон (transliteration)	DB01118	approved, investigational	small molecule
6	digoxin	дигоксин (transliteration)	DB00390	approved	small molecule

Identifiers: DrugBank. INN: DrugCentral, joined on InChIKey (no value means no match). Russian names: rule-based transliteration — an interface label, not sourced data.

3 Pairwise map



Lines: solid — database or label record; dashed — model signal only; dotted — no documentation found. A thick line marks a pair flagged for a closer look.

Pair matrix

Each cell is the kind of documentation found for the pair (row × column); the number is the calibrated model score. Colour encodes the kind of source, not danger. “n/c” — not counted: the model was trained on this pair's record.

	1	2	3	4	5	6
1 Warfarin		DB 0.81	DB 0.48	DB 0.49	DB 0.83	– 0.45
2 Fluconazole			DB n/c	DB 0.91	DB n/c	DB n/c
3 Simvastatin				DB 0.84	DB n/c	M 0.58
4 Clarithromycin					DB 0.98	DB 0.53
5 Amiodarone						DB n/c
6 Digoxin						

CI Contraindication in a label **L** Named in a label **≠** Sources conflict **DB** Documented in a database **M** Model signal only
– No documentation found

4 All pairs: evidence and model

For each pair the service reports the number of mechanism hypotheses; the hypotheses, proteins and pathways themselves are in the per-pair report (Analyzer).

PAIR	WHAT WAS FOUND	DRUGBANK	LABELS	HYP.	MODEL	WHY LOOK CLOSER
Warfarin + Fluconazole 1–2 · DB00682 · DB00196	Documented in a database	1	0	3	0.811 S2	database record; pharmacokinetic mechanism hypothesis
Warfarin + Simvastatin 1–3 · DB00682 · DB00641	Documented in a database	1	0	4	0.484 S2	database record; pharmacokinetic mechanism hypothesis
Warfarin + Clarithromycin 1–4 · DB00682 · DB01211	Documented in a database	1	0	2	0.489 S3	database record; pharmacokinetic mechanism hypothesis
Warfarin + Amiodarone 1–5 · DB00682 · DB01118	Documented in a database	1	0	7	0.832 S2	database record; pharmacokinetic mechanism hypothesis
Fluconazole + Simvastatin 2–3 · DB00196 · DB00641	Documented in a database	1	0	3	not counted S1	database record; pharmacokinetic mechanism hypothesis
Fluconazole + Clarithromycin 2–4 · DB00196 · DB01211	Documented in a database	1	0	3	0.914 S2	database record; pharmacokinetic mechanism hypothesis
Fluconazole + Amiodarone 2–5 · DB00196 · DB01118	Documented in a database	1	0	2	not counted S1	database record; pharmacokinetic mechanism hypothesis
Fluconazole + Digoxin 2–6 · DB00196 · DB00390	Documented in a database	1	0	1	not counted S1	database record; pharmacokinetic mechanism hypothesis
Simvastatin + Clarithromycin 3–4 · DB00641 · DB01211	Documented in a database	1	0	5	0.841 S2	database record; pharmacokinetic mechanism hypothesis
Simvastatin + Amiodarone 3–5 · DB00641 · DB01118	Documented in a database	1	0	5	not counted S1	database record; pharmacokinetic mechanism hypothesis
Clarithromycin + Amiodarone 4–5 · DB01211 · DB01118	Documented in a database	1	0	5	0.979 S2	database record; pharmacokinetic mechanism hypothesis
Clarithromycin + Digoxin 4–6 · DB01211 · DB00390	Documented in a database	1	0	5	0.530 S2	database record; pharmacokinetic mechanism hypothesis
Amiodarone + Digoxin 5–6 · DB01118 · DB00390	Documented in a database	1	0	3	not counted S1	database record; pharmacokinetic mechanism hypothesis
Simvastatin + Digoxin 3–6 · DB00641 · DB00390	Model signal only	0	0	5	0.576 S1	model signal without documentation; pharmacokinetic mechanism hypothesis
Warfarin + Digoxin 1–6 · DB00682 · DB00390	No documentation found does not mean “safe”	0	0	0	0.454 S2	—

5 Coverage and uncertainty

- The regime says whether the pair's drugs were in the model's training set: S1 — both (not evaluated), S2 — one, S3 — neither.
- For each pair the service reports the number of mechanism hypotheses; the hypotheses, proteins and pathways themselves are in the per-pair report (Analyzer).
- The service uses one model checkpoint (seed 0) and provides neither a per-pair confidence interval nor an ensemble estimate. Measured performance for this regime is given instead — an average over many pairs, not confidence in this pair.

Pairs whose score is not counted

- Fluconazole + Simvastatin (S1)
- Fluconazole + Amiodarone (S1)
- Fluconazole + Digoxin (S1)
- Simvastatin + Amiodarone (S1)

- Amiodarone + Digoxin (S1)

Sources checked

SOURCE	STATUS	DETAIL
DrugBank (TDC export, local)	checked	Local DrugBank export (TDC).
FDA labels (openFDA)	unavailable	openfda_label: recently unreachable; retrying after a cooldown

6 Sources and reproducibility

Endpoint (multi)	http://127.0.0.1:8765/api/v3/multi
Request body (multi)	{"drugs":["DB00682","DB00196","DB00641","DB01211","DB01118","DB00390"],"include_labels":true}
Response received by the browser (multi)	2026-10-15T04:53:34.101Z
Report generated by the service (generated_at)	2026-10-15T04:53:34+00:00
API report version	3.0
Model	BIO-GINE M4 — frozen checkpoint bd45f84e3c1b2c33 per the site's frozen data; the multi-drug response itself does not name it
Frozen research tag	v2-final-github-safe-2026-09-03 · 92c481eeaba8faff991ced850e1c4de418ea31b0
Site	http://127.0.0.1:4173
Site build commit	not recorded
PDF generator	DDI-Net web, in-browser export (@react-pdf/renderer)
Generated	15 October 2026, 04:53:34 (UTC, UTC+00:00)
Generated (UTC)	2026-10-15T04:53:34.361Z
Language	English
SHA-256 of the response data	a6c3c708b3c4d60af34599dc0291e6b4f7c08f0ec85f33610c0746919b97d004 Hash of the service response(s) as canonical JSON (keys sorted, serialised with JSON.stringify). The same hash is in the companion JSON file, available from the “JSON” button next to “Export PDF”.

7 Limitations

Core limitations

- DDI-Net is a research tool, not clinical decision support and not a medical device. Decisions about medicines belong to a doctor or pharmacist.
- Unlabelled pairs are not confirmed safe. A missing record means only that the sources checked do not contain one.
- BIO-GINE scores are not calibrated probabilities of clinical harm: temperature scaling was fitted on validation pairs at 50% prevalence against randomly drawn unlabelled pairs.
- The drug order in the DrugBank (TDC) export was verified by measurement for only two interaction types; for the others it is either meaningless (symmetric types) or unverified, so directional claims (“A raises the level of B”) are not made.
- External medical evidence (FDA labels, RxNorm) may be incomplete or unavailable at query time; its status is listed under “Sources checked”.
- Pairwise results say nothing about three or more drugs taken together; no such model exists here, and no overall score is computed.

RESEARCH USE ONLY

This is a research prototype, not a medical device. It does not replace a doctor or pharmacist and cannot say whether a combination is safe for a particular person.

A Appendix A. Machine-readable summary (JSON)

Key fields as JSON for automated checking. The full service responses are in the companion JSON file (same SHA-256).


```
{
  "schema": "ddinet.pdf-summary/1",
  "report_id": "DDI-20261015-045334-A6C3C708",
  "generated_at": "2026-10-15T04:53:34+00:00",
  "generated_at_utc": "2026-10-15T04:53:34.361Z",
  "time_zone": "UTC",
  "language": "en",
  "payload_sha256": "a6c3c708b3c4d60af34599dc0291e6b4f7c08f0ec85f33610c0746919b97d004",
  "inputs": [
    {"endpoint":"http://127.0.0.1:8765/api/v3/multi","request":{"drugs":["DB00682","DB00196","DB00641","DB01211","DB01118",
      "DB00390"],"include_labels":true},"received_at":"2026-10-15T04:53:34.101Z"}
  ],
  "server_generated_at": "2026-10-15T04:53:34+00:00",
  "report_version": "3.0",
  "drugs": [
    {"id":"DB00682","inn":"warfarin","name_ru":"варфарин"},
    {"id":"DB00196","inn":"fluconazole","name_ru":"флуконазол"},
    {"id":"DB00641","inn":"simvastatin","name_ru":"симвастатин"},
    {"id":"DB01211","inn":"clarithromycin","name_ru":"кларитромицин"},
    {"id":"DB01118","inn":"amiodarone","name_ru":"амиодарон"},
    {"id":"DB00390","inn":"digoxin","name_ru":"дигоксин"}
  ],
  "pairs": [
    {"a":"DB00682","b":"DB00196","status":"documented_no_severity","db":1,"labels":0,"hypotheses":3,"score":0.8108211344965719,
      "regime":"S2","label_seen":false,"review":true},
    {"a":"DB00682","b":"DB00641","status":"documented_no_severity","db":1,"labels":0,"hypotheses":4,"score":0.48439914908419174,
      "regime":"S2","label_seen":false,"review":true},
    {"a":"DB00682","b":"DB01211","status":"documented_no_severity","db":1,"labels":0,"hypotheses":2,"score":0.4894444855664803,
      "regime":"S3","label_seen":false,"review":true},
    {"a":"DB00682","b":"DB01118","status":"documented_no_severity","db":1,"labels":0,"hypotheses":7,"score":0.8323730133899702,
      "regime":"S2","label_seen":false,"review":true},
    {"a":"DB00682","b":"DB00390","status":"no_documented_evidence_found","db":0,"labels":0,"hypotheses":0,"score":
      0.45442367764364605,"regime":"S2","label_seen":false,"review":false},
    {"a":"DB00196","b":"DB00641","status":"documented_no_severity","db":1,"labels":0,"hypotheses":3,"score":null,"regime":"S1",
      "label_seen":true,"review":true},
    {"a":"DB00196","b":"DB01211","status":"documented_no_severity","db":1,"labels":0,"hypotheses":3,"score":0.9137336264872571,
      "regime":"S2","label_seen":false,"review":true},
    {"a":"DB00196","b":"DB01118","status":"documented_no_severity","db":1,"labels":0,"hypotheses":2,"score":null,"regime":"S1",
      "label_seen":true,"review":true},
    {"a":"DB00196","b":"DB00390","status":"documented_no_severity","db":1,"labels":0,"hypotheses":1,"score":null,"regime":"S1",
      "label_seen":true,"review":true},
    {"a":"DB00641","b":"DB01211","status":"documented_no_severity","db":1,"labels":0,"hypotheses":5,"score":0.8411772377234219,
      "regime":"S2","label_seen":false,"review":true},
    {"a":"DB00641","b":"DB01118","status":"documented_no_severity","db":1,"labels":0,"hypotheses":5,"score":null,"regime":"S1",
      "label_seen":true,"review":true},
    {"a":"DB00641","b":"DB00390","status":"model_signal_only","db":0,"labels":0,"hypotheses":5,"score":0.5755806684194835,"regime":
      "S1","label_seen":false,"review":true},
    {"a":"DB01211","b":"DB01118","status":"documented_no_severity","db":1,"labels":0,"hypotheses":5,"score":0.9786644442349784,
      "regime":"S2","label_seen":false,"review":true},
    {"a":"DB01211","b":"DB00390","status":"documented_no_severity","db":1,"labels":0,"hypotheses":5,"score":0.5295242151920286,
      "regime":"S2","label_seen":false,"review":true},
    {"a":"DB01118","b":"DB00390","status":"documented_no_severity","db":1,"labels":0,"hypotheses":3,"score":null,"regime":"S1",
      "label_seen":true,"review":true}
  ],
  "sources_checked": [
    {"source":"drugbank_tdc","status":"ok"},
    {"source":"openfda_label","status":"unavailable"}
  ]
}
```